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Slim ESP Performance in Challenging Applications with Y-Tool, PMM, and High-Speed Pumps

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Abstract

This paper illustrates the benefits of Slim series ESP applications for challenging wells with different type of limitations such as well construction, reservoir data and type of installation for various wells in UAE.

The multitasks of Slim series ESP allows to use this application for deviated wells with 7" liner casings with weight 29# where well access is mandatory for wireline/slickline and coiled tubing operations.

In addition, the design of Slim series ESP allows to install equipment as much as possible with inclination angle up to 90 degrees where wells have low static pressure or tendency to decrease static pressure.

Slim series ESP is installed with Permanent magnet motor (PMM) and allows to operate up 6000 RPM. Also, it is possible to reach flowrate up to 4000 BPD.

The solution was to install Slim series ESP + Y-tool inside liner casing 7" #29. It allows slickline/wireline and coiled tubing with standard tools to reach the reservoir. More than 15 wells equipment were installed for several onshore fields in UAE. The first installed ESP is still operating with current run life more than 500 days.

the Slim series ESP is appropriate solution for deviated wells where 7" liner has limited space for ESP + Y-tool installation.

Additionally, Slim series pump design with Flush Joint Threaded (FJT) connection allows to reach flow rate up 4000 BPD in such small equipment size 3.19 inches without any additional friction losses compare with standard flange connection.

These key factors allow to cover most of onshore oil wells in UAE.

The variability of Slim series ESP allows to cover different challenging production wells cases and reach target data. Also, all Slim series ESP strings are equipped with high efficiency and high-speed permanent magnet motors. It helps to save energy, decrease CO2 footprint and survive in harsh well conditions.

The Features of Slim series ESP Technology

The Slim series ESP technology includes a range of design features and engineering solutions that enable this equipment to effectively meet a wide spectrum of production requirements, particularly in complex and challenging well environments. These features make it suitable for wells with demanding operational

parameters, where standard ESP systems may face performance or installations limitations. Specially, the Slim series ESP is capable to handling production flow rates, with capacity reaching up to 4000 BPD, which is especially important for high-output wells with restricted casing sizes. Additionally, the system is compatible with Y-tool configurations, allowing for dual access to the wellbore either for simultaneous ESP operation and intervention activities or for monitoring and chemical injection purposes. Moreover, the Slim series ESP is well-suited for installation in highly deviated or horizontal wells, where compact equipment with robust mechanical performance and flexible installation geometry is essential. These design capabilities ensure reliable and efficient operation under a variety of well conditions, making the Slim series ESP a versatile and adaptation solution for modern oilfield applications.

The pump features a unique stage design with improved geometric accuracy, ensuring consistently high performance when handling multiphase fluids such as oil, water and gas. This precision-engineered configuration allows the pump to reach efficiency levels up to 70%, which is particularly notable for a compact 3.19-inch diameter system (Figure 1). Such efficiency is rarely achieved in equipment in of this size and is the result of extensive design optimization and material selection aimed minimized hydraulic losses. Thanks to these design enhancements, the Slim series ESP can operate at a nominal speed of up to 6000 RPM. For demanding scenarios requiring flow rates above 4000 BPD, the system is capable of running at speeds up to 9000 RPM while maintaining operational stability, mechanical reliability and extended run life, even in challenging wellbore conditions with fluctuating pressures and fluid compositions.



Figure 1—Slim series ESP stage

One key feature of the updated pump design is its capability to handle high flow rates up to 4000 BPD. In conventional ESP systems, pump sections are typically connected using flange connections, which, especially in small diameter housings (3.19 inches), create internal flow restrictions. These restrictions become significant at flow rates above 2300 BPD due to reduced internal clearance and abrupt transitions between sections. The redesigned connection used in the Slim series ESP minimizes hydraulic losses by providing a smoother internal flow path. This allows for efficient operation at higher flow rates without the performance drop typically associated with flange connections in compact systems (Figure 2).

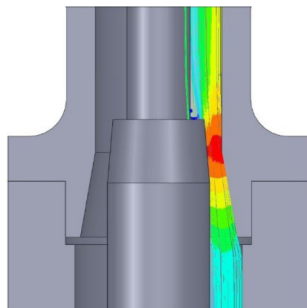


Figure 2—Local resistance and losses in flowing channel

The design of the pump connection was revised in order to eliminate increased hydraulic losses commonly associated with conventional flange-type connections. These losses can become particularly critical when at elevated flow rates, where even minor restrictions or discontinuities in internal flow paths may significantly impact overall system efficiency and performance. To address this issue, The Flush Joint Threaded (FJT) connection was implemented as an optimized solution specially tailored for high-flow rate applications. The FJT connection offers a streamlined internal profile, which minimizes turbulence and pressure drop at the connection interfaces. By providing a smoother transition between adjacent components, the design reduces energy losses and improves hydraulic efficiency across the entire pump assembly. This enhanced connection method was especially beneficial for the Slim series ESP system, where space limitations and compact design require precise alignment and minimal flow path obstruction. The introduction of the FJT connection allowed for the effective elimination of choke points and flow restrictions that previously existed between various mechanical components of the ESP string (Figure 3).

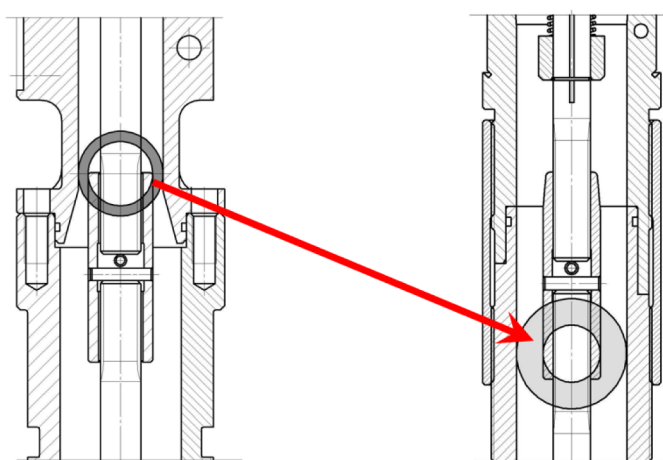


Figure 3—From flange connection to Flush Joint Threaded (FJT)

Additionally, FJT (Flush Joint Thread) connection incorporates several specific design features that enhance its performance are reliability during extended operational periods in downhole environments. A key component of this configuration is the threaded sleeve located at the head/base of the pump and head of the intake, providing a robust and leak-resistant connection capable of withstanding high axial and torsional loads during operation. To further improve the precision and repeatability of the assembly process, a special alignment pin is integrated into the pump/intake head/base part. Correspondingly, the dedicated grooves are machined into the head/base. These grooves are designed to accept the alignment pin and ensure proper orientation equipment during installation. Importantly, either of the two grooves can be utilized during assembly, which offer flexibility in the field operations. The only requirement is that Motor Lead Extension (MLE) must be correctly positioned over the reference area on the pump base (Figure 4).

Slim series ESP systems offer a significant competitive advantage when it comes to installation and reliable operation in wells characterized by high dog leg severity (DLS). Conventional ESP systems are generally limited to installation at maximum inclination angles up to 60 degrees. The restriction is primarily due to the design limitations of the labyrinth chambers within the seal section and rotor-stator alignment constrains in the motor, which are not optimized for high-deviation applications. By comparison, Slim series ESP is engineered with more flexible configuration. The protector section utilizes only bag-type chambers, eliminating the dependence on labyrinth-type seals. As a result, the system can accommodate installation at inclination angles of up to 90 degrees at the ESP installation depth without compromising the performance or integrity of the system components. Furthermore, Slim series ESPs are commonly equipped with Permanent Magnet Motors (PMM), which are designed to operate efficiently in horizontal or highly deviated wellbores.

The PMM in this configuration features a universal design capable of withstanding inclination angles up to 90 degrees. The capability is achieved through a simplified yet effective modification: in inclusion of a specialized upthrust bearing within the motor assembly. Thi single, compact design element allows the motor to function reliably regardless of deviation, effectively making it a universal solution for high-angle installations.

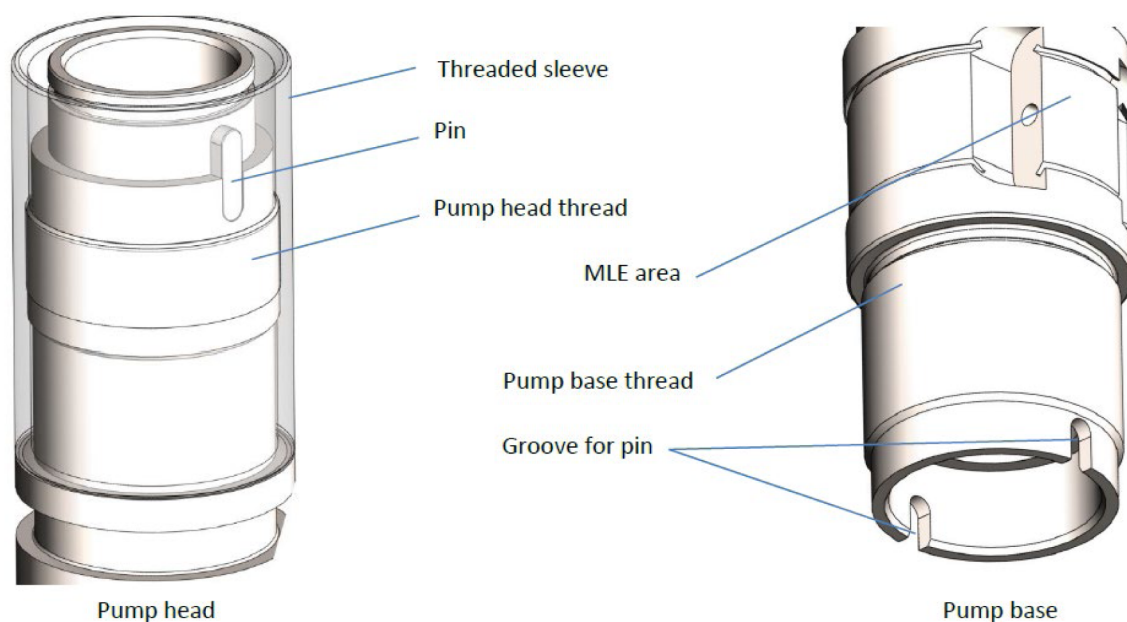


Figure 4—Flush Joint Threaded (FJT) connection

Additionally, the motor design has been optimized to support stable operation at frequencies up to 9000 RPM. Design improvements include enhanced rotor dynamics and thermal management to accommodate increased rotational speeds. Rotor packs now feature higher-precision dynamic balancing, reducing residual unbalance and associated vibrations. Manufacturing tolerances have been tightened to ensure consistent mechanical integrity and minimize deviations affecting high-speed performance.

Production Challenges

In the heart of the Middle East, a leading energy oil producer faced a set of technical and operational challenges in one of its long- producing onshore fields. Over the years, the natural decline in static reservoir pressure had eroded the effectiveness of existing artificial lift systems, making it increasingly difficult to sustain historical production rates. At the same time, the operator needed to maintain full reservoir accessibility for critical interventions such as wireline and slickline operations, as well as periodic well stimulations.

Adding to the complexity, certain wells exhibited integrity issues with the tightness of their production casings. These factors demanded a completion strategy that could restore production performance while ensuring long-term well integrity and operational flexibility.

After careful evaluation, the engineering team identified two potential approaches: installing a tie-back string of new production casing with a smaller inner diameter, or drilling a sidetrack completed with the sane dimensions. In both scenarios, the optimal design involved 7-inch casing either 26 or 29 lbs/ft offering the required structural strength while working within wellbore constraints.

The company's production objectives were clear and ambitious: achieve up to 1000 BPD from one reservoir zone and as much as 2000 BPD from another. However, the reduced casing diameter left little

room for conventional equipment, especially given the requirement to run standard logging tools without obstruction

The solution came in the form of Slim series ESP technology paired with a Y-tool assembly. This innovative configuration fits seamlessly within the 7-inch casing, provides unentrapped access for well interventions, and delivers the lift performance needed to reach the targeted production rates (Figure 5).

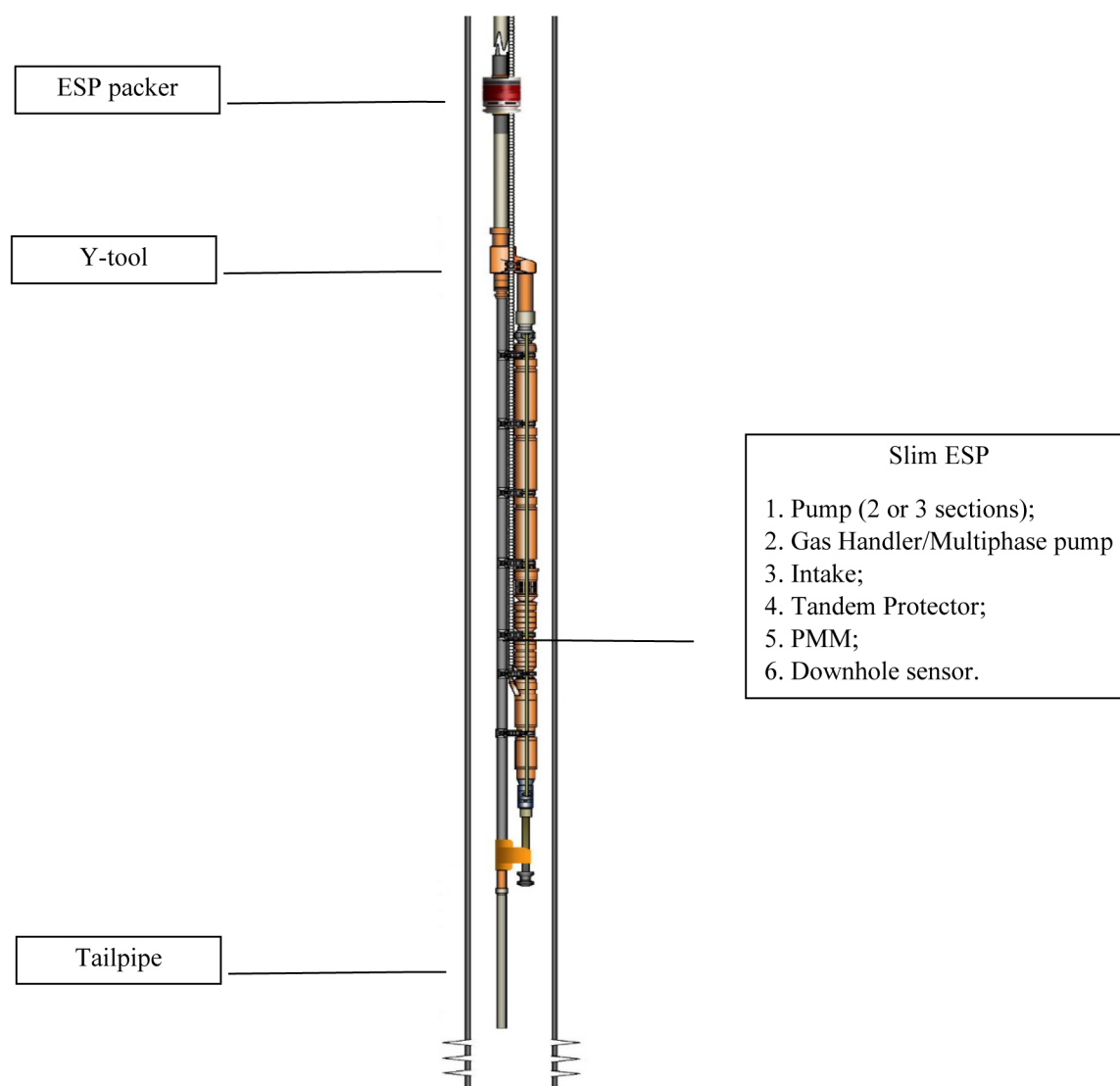


Figure 5—Slim ESP with Y-tool

By adopting this customized approach, the operator successfully met both its production and operational goals. The project stands as a prime example of how advanced, fit-for-purpose technology can overcome space, pressure and accessibility constraints in mature oilfield- delivering sustainable output and preserving intervention flexibility for years to come.

Implementation

To meet the ambitions production targets from multiple reservoir zones, the operator selected two of Slim series ESPs, each with an outer diameter of just 3.19 inches. These compact yet powerful pumps were chosen specifically to deliver the required flow rates from sperate intervals while operating within the tight dimensional constraints of the completion.

A key enabling component was a custom-engineered Y-tool designed for compatibility with 7-inch 32# casing. This casing specification has a drift I.D. of 5.969 inches (Figure 6), and the maximum O.D. of entire ESP system - including ESP bypass clamp - was carefully kept smaller than this drift I.D. to ensure trouble-free passage. To verify dimensional compatibility and eliminate installation risks, the project team conducted a full stack-up test prior to the first field deployment. This procedure confirmed that both Slim series ESP and Y-tool assembly could pass smoothly through the casing and operate without interference.

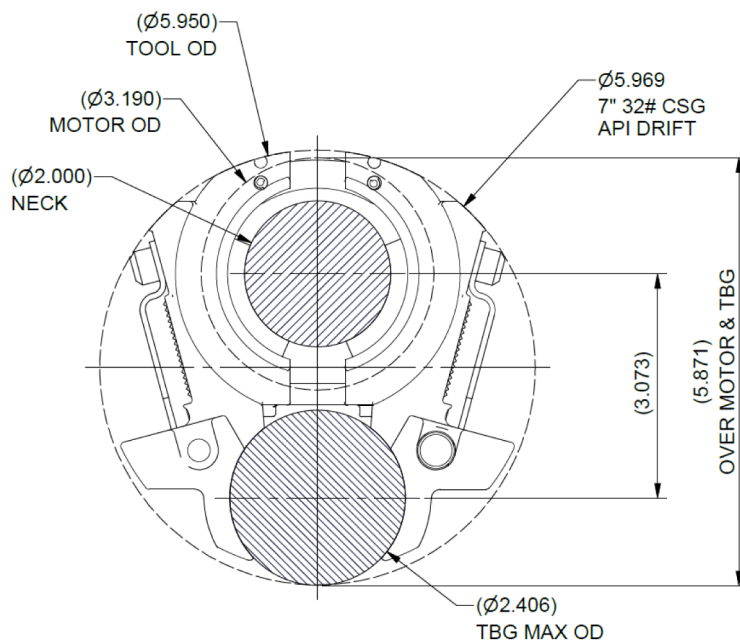


Figure 6—Cross-section Slim series ESP + Y-tool

Given the continuing decline in static reservoir pressure, long term operational stability was a critical design consideration. To mitigate gas lock one of the most common causes of premature ESP shutdown in low-pressure wells a specialized handle device was integrated into each pump string. This measure was intended to maintain consistent performance as wellbore conditions evolved over time. The design target for this installation was an impressive run life approximately 2000 days, reflecting the company's focus on reliability and lifecycle cost optimization.

The pump assembly is built with a compression design, ensuring mechanical integrity and operational stability even under challenging downhole conditions. When combined with Permanent magnet Motor (PMM), the system is capable of operating over a wide frequency range from 40 Hz to 100 Hz providing exceptional flexibility in adapting to varying production requirements (Figure 7). This broad operational window allows the ESP to respond effectively to fluctuations in well inflow that may occur due to unexpected changes in reservoir pressure, fluid properties and/or reservoir productivity. An additional and equally important advantage of the Slim series ESP comes from its use of the PMM itself. This technology proved particularly valuable in this project, where available power is both limited and subject to strict monitoring to ensure efficient energy use. The PMM's high efficiency and reduced head losses mean that less electrical energy is wasted during operation, thereby maximizing the amount of input power converted into useful pumping work. From a practical standpoint, the implementation of PMM technology has resulted in significant energy savings, lowering the total power consumption by up to 30% per ESP system compared to conventional ESP systems with induction motors. These savings translate into not only reduced operational costs but also a measurable decrease in the environmental footprint of production activities supporting both the company's goals and its sustainability objectives.

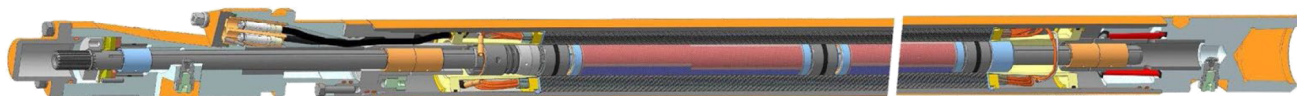


Figure 7—Permanent Magnet Motor (PMM)

As part of completion, an ESP packer was installed to perform dual functions, protecting the new production casing from potential fluid and pressure damage, and serving as a robust mechanical barrier within the well. The Slim series ESP distinctive design offers a significant advantage in terms of wellbore navigability. Unlike conventional ESP systems - many of which are limited to inclinations of about 60 degrees due to protector design Slim series ESP configuration is capable of passing through and operating in sections with high dog-leg severity, even in wells inclined up to 90 degrees.

The enhanced flexibility allowed the operator to position the new Slim series ESP deeper in the well than the previous equipment installation. The result was a marked improvement in performance, including higher sustained flow rates, greater operational stability, and the ability to the pump intake above the bubble point pressure. By doing so, the system minimized gas breakout in the pump and ensured a more consistent liquid flow to surface.

Additionally, there is one more critical consideration before installing the Slim series ESP equipped with a Permanent Magnet Motor (PMM) ensuring full compatibility with the existing surface equipment. If the current SKID is fitted with a Fix Drive, the entire SKID must be replaced with a Variable Speed Drive (VSD) capable of operating with both PMM and AM. Such a solution provides a longterm operational advantage, as it allows for future ESP reinstallation with an induction motor without additional surface equipment changes. Another common scenario is when the existing SKID already has a VSD, but the unit is not designed to operate with PMM technology. In this case, the recommended solution is to retrofit only the VSD, replacing it with a universal model compatible with both motor types. This approach not only ensures seamless integration and reduces installation downtime but also minimizes replacement costs while increasing operational flexibility for future field requirements (Figure 8).

			Start		Stop	
Field-1 cluster-1 well-1			10:57:16 16.02.2018			
<			STATUS			
VSD mode		Auto	STARTING			
Output frequency		0.00 Hz				
RPM Motor		0 Rpm	In work	0:00:02		
Motor current		0.0 A	Motor type	PMM		
Loading		0 %	Control type	U/f		
Power factor		0.00	Rotation	Reverse		
Insulation Resist		9999 kOhms	Start-up method	Smooth		
Intake Pressure		---	Technological regimes			
Motor Temp.		---				
To move use arrows						

Figure 8—Universal VSD controller for operating both PMM and AM

Overall, the deployment of the Slim series ESP system, combined with the purpose-build Y-tool, represents a tailored solution that overcome significant space and inclination constraints, extended equipment run life, and unlocked higher production potential from oil fields.

Outcomes and Future actions

To date, a total of fifteen Slim series ESP systems, each with an O.D. of just 3.19 inches and equipped with custom-designed Y-tools, have been successfully installed across multiple wells in the field. The deepest pump setting depth achieved so far reaches nearly 7300 feet, showcasing the system's capability to operate reliably at significant depths. In terms of trajectory, the installations have also proven their ability to function in challenging wellbore geometrics, with the maximum inclination angle recorded at approximately 80 degrees a level of deviation that would be problematic for conventional ESP systems.

The performance improvements have been notable. In several wells, the ability to install pump deeper than previous configurations has resulted in production rates doubling, significantly boosting oil recovery. The deeper replacement not only enhanced fluid extraction but also ensured that the pump intake pressure remained above the bubble point pressure, promoting stable and continuous operation. In other wells, while the gains were more moderate, production still increased 10-15%, contributing to the overall field uplift.

All installations were powered by Permanent magnet motors (PMMs), which played a key role in improving energy efficiency. The high-efficiency PMM design directly translated into lower operating costs, reduced power consumption, and a smaller CO₂ footprint compared to conventional induction motors. From an economic stand point, the technology also delivered measurable benefits by reducing the cost per barrel, and outcome driven both by motors efficiency and the resulting production gains.

This combination of advanced ESP design, flexible deployment capability, and high-efficiency motor technology has demonstrated not only strong technical performance but also a clear advantage, reinforcing the value of adopting fit-for-purpose artificial lift solutions in mature, complex wells.

Looking ahead, the future deployment plans for Slim series ESP combined with Y-tool assembly are extensive, covering all of the operation's oilfields where production requirements exceed 2000 BPD and whole operating conditions present greater technical challenges. The challenges include more demanding reservoir environments, such as higher temperature and variable fluid properties, which place additional stress in artificial lift equipment. In such harsh conditions, the use of Permanent Magnet Motor (PMM) is expected to deliver significant benefits in terms of reliability and extended run life. According to the NEMA (National Electrical Manufacturers Association) standard, motor insulation life is reduced by half for every 10°C increase in operating temperature. This means that in high-temperature reservoirs, selecting a motor technology capable of maintaining efficiency and minimizing thermal stress is essential. The PMM's inherently higher efficiency results in low heat generation compared to conventional induction motors, which in turn helps preserve the integrity of the motor insulation over a longer operational period.

However, achieving the desired performance at higher flow rates will require design refinements to the Y-tool assembly, particularly in the bypass clamps. In current configuration, standard bypass clamps are not well-suited for high-rate applications because they lack the necessary geometry specifically, there is no 'neck' between pump sections in Flush Joint Threaded (FJT) connection. To address this limitation, the bypass clamp mechanism will be redesigned with a frictional holding system that secures the clamp between the ESP and the bypass tubing without requiring a traditional neck profile. This friction-based design will ensure a secure fit while still allowing for straightforward assembly during installation. Additionally, the improved clamp design will help maintain the structural integrity of the bypass assembly under the higher mechanical loads and fluid velocities associated with increased production rates.

By implementing these planned modifications and continuing to leverage the efficiency and durability of PMM technology, the Slim series ESP + Y-tool configuration will be capable of meeting the diverse production demands across the operation's portfolio delivering both high performance and extended operational life in even the most challenging reservoir conditions.

References

Ultra Slim ESP - Extending ESP Application (Paper Number: SPE-221534-MS)